Fiscal Year:	FY 2024	**Task Last Updated:**	FY 05/16/2024
PI Name:	Wang, Zheng Ph.D.		
Project Title:	Investigating the Roles of Melanin and DNA Repair on Adaptation and Survivability of Fungi in Deep Space		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	United States Naval Research Laboratory	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Washington	**State:**	DC
Zip Code:	20375-0001	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-Artemis1 (EM1). App A: Orion (Artemis-1) (formerly Exploration Mission-1)
Start Date:	07/15/2019	**End Date:**	10/15/2023
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Freeland, Denise	**Contact Phone:**	321-867-5878
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Flight Program:			
Flight Assignment:	End date changed to 10/15/2023 per D. Freeland/KSC (Ed., 4/30/2024) End date changed to 07/15/2023 per H. Levine/KSC (Ed., 2/27/23)		
Key Personnel Changes/Previous PI:	Dr. Jennifer Yuzon left the Naval Research Laboratory (NRL) in February, 2023. Tiffany Hennessa, Ph.D. (formerly Dr. Tiffany Wong-Stack) remains as Co-Investigator.		
COI Name (Institution):	Hennessa, Tiffany (Naval Research Laboratory)		
Grant/Contract No.:	NNK19OB09A IAA		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight to regions beyond low Earth orbit involves exposure to unique environmental hazards, most notably radiation. Humans traveling to these regions will be exposed to radiation from cosmic rays, which will cause DNA damage and oxidative stress. The extent of this damage, however, is unknown, and more basic research into the genetic effects of combined cosmic ray and microgravity exposure is needed. We propose to study these effects using a type of organism that is almost certain to inadvertently accompany astronauts on all of their flights – fungi. Fungi are remarkably stress-resistant and have been isolated several times from spacecraft. The objective of this proposal is to use the well-studied mold Aspergillus nidulans to test two mechanisms for space adaptation – DNA repair and the production of melanin. Mutants of this organism that cannot make melanin or are defective in a type of DNA repair associated with recovering from radiation will be used. When the samples return, the spores that these strains produce will be tested for their survival, and their protein and DNA will be analyzed to find molecular signatures of adaptation to deep space. The data we collect will tell us several things: whether melanin protects from the stresses faced during spaceflight, the types of DNA damage that occur in space, and what changes occur in fungi when they are adapted to prolonged spaceflight. The results will determine characteristics of the fungi that astronauts may have to control on future missions, including pathogenic species. Because fungi share many proteins with humans, the data collected on the proteins involved in adaptation can be applicable to astronaut health. Finally, determining whether melanin assists with survival in space will provide more evidence for it to be used as a protective material for several future NASA applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project focuses on the tools that fungi use to be resilient against stress. In this case, the stress of deep space, which consists of a unique composition of radiation and microgravity that has rarely, if ever, been experienced by humans. However, taking astronauts to locations in space far beyond the protection of Earth's atmosphere are aspirations of all space programs, and the stress that is associated with deep space radiation exposure (for instance, free-radical damage) overlaps in many of its biological effects with more routine stresses faced on earth. Fungi, which thrive in extreme environments such as space, and possess a genetic makeup that is similar to humans but also much simpler, are also an ideal group of organisms for understanding and combating radiation and other stresses. This project addresses two components of stress resistance in the fungus Aspergillus niger – the production of melanin and DNA repair proteins, to determine the extent and nature of their importance in survival, adaptation, and damage protection during an extended flight in deep space. The data collected from the Artemis flight will inform our understanding of how cells and organisms resist, or can be made to resist, the stress of space and other damaging environments.
Task Progress:	February 2025 Update from the Principal Investigator (PI): The investigation has been ended. We are currently preparing a manuscript for this project (Ed., 2/7/25) Year 5 Progress Ground control experiments were scheduled on April 3, 2023 asynchronously from the flight experiment. Spores were prepared in the same manner as the experimental set and shipped to the NASA Kennedy Space Center for experimental simulation. Afterwards, samples were returned on May 23, 2023 to the U.S. Naval Research Laboratory in Washington, D.C. Once samples were received, tubes were removed and single spores were selected and spread onto solid Yeast Extract–Peptone–Dextrose (YPD) media 30 °C for 24-48 h and spread onto plates again to ensure single colonies. The remaining spores were collected with sterile water and transferred into tubes with equal volume of 30% glycerol in cry-vials and stored at -80 °C. Spore Survivability Germination analysis was performed by diluting a portion of the 1 mL spore suspensions (1:100) with 100 µL aliquots spread onto YPD plates for 24 h. Germination was read from 100 spores under a microscope at 20 x objective. Germination rates were calculated as a percentage and is highly indicative of survivability. Spore survival in the ground control samples showed the highest range of variation, compared to the flight set where that was more consistent. Nonetheless, spore survival was high in both the ground controls and flight samples. Fungal Biomass Biomass was calculated by removing fungal tissue from Pluronic medium by incubating at 4 °C for at least 2 h until liquefied. The fungal tissue was pelleted by centrifugation at 10,000 g for 10 minutes 4 °C. Fungal pellets were washed with ddH2O and flash frozen in liquid nitrogen and subsequently lyophilized for 48 h. Lyophilized pellets were weighed and analyzed between the ground control and flight samples. Biomass was found to be greater in FWN, KUS, and UVSC strains compared to the respective ground control samples with the most variation in the DNA repair defective strains (KUS and UVSC). Spore Morphology Spores from single colonies were collected and imaged using oil-immersion 63x Differential Interference Contrast (DIC) objective on the Zeiss AxioObserver inverted microscope equipped with 385 nm, 630 nm, and transmitted light LEDs. Brightfield images were used to measure spore diameters. Upon opening the image in ImageJ, clumped cells and out of focus cells were removed and a binary mask was created using the auto threshold function. The mask was saved and processed using an in-house MATLAB code. The distance between each object and every other object was calculated and any object with a minimum distance less than or equal to the minimum allowed distance (10 pixels) was removed. Lastly, objects were filtered by their circularity, keeping objects with a circularity of 0.85 or greater. Differences in spore diameters were found between: wild type (WT) ground control (GC) and flight (FL) sets, WT GC and KUS GC, and WT GC and UVSC GC. Currently scanning electron microscopy (SEM) imaging is being performed to understand morphological changes. Radiation Profile Radiation profiles were characterized by using area monitoring Crew Active Dosimeters (CAD) from March 3, 2023 to July 6, 2023 for the asynchronous ground control experiment, and October 4, 2022 to December 11, 2022 for the flight experiment. CADs have a sensitivity/resolution of 20 µGy and can display daily cumulative doses and rate of radiation. Genomic Sequencing

For genomic analysis of single nucleotide polymorphisms, A. niger single colony plates were grown in 60 mm petri dishes in liquid YPD media and incubated for five days at 30 °C. Fungal tissue was collected and flash frozen in liquid nitrogen. DNA was isolated with cetyltrimethylammonium bromide (CTAB) reagent and purified and sequenced at the Yale Center for Genomic Analysis and performed on the Illumina NovaSeq 6000 sequencing platform using the paired-end method. Sample alignment to screen for genetic variants is performed by using FastQC, paired end reads were trimmed off using Trimmomatic and mapped to the A. niger genome ASM285v2. Variants were called using BCFtools and used to generate new reference genomes using the Genome Analysis Toolkit (GATK3). Currently, genomic variant analysis is still ongoing.

Metabolomic Analysis

Metabolomic analysis to profile and compare metabolites in biological samples are currently ongoing. The current metabolomic approach using High Performance Liquid Chromatography (HPLC) involved using a Thermo Scientific U3000 instrument using an ultraviolet (UV) diode array detector (DAD) for absorbance of untargeted metabolites in each sample. Liquid Chromatography (LC) data is currently being analyzed in Chromeleon software and compared between ground control and experimental flight samples.

Summary

We report that our fungal samples have successfully orbited the Moon on the Artemis I mission in the Bio-Exp1 program. We have also completed the asynchronous ground control experiment at KSC that closely mimics the duration of the flight. We are currently summarizing our data involving spore survivability, fungal biomass, morphology, and radiation profiling. Our work continues with determining genomic variation and identifying untargeted metabolites between our samples.

Bibliography Type: Description: (Last Updated: 06/06/2023)